SAKHI

Product Requirements Document

A private, multilingual women’s health companion designed to listen first, understand the user, and provide safe, evidence-grounded guidance.

Product Type	Women’s Health AI Companion
Primary Users	Girls and Women
Core Experience	Empathetic conversation + personal health tools + safe referral
Document Purpose	Single source of truth for the product scope and requirements
Version	1.0

Product principle: The user should feel that she is talking to an understanding companion, not filling out a medical form or searching the internet.

1. Executive Summary

Sakhi is a private women's health companion for girls and women. The product combines a simple personalized health dashboard with an AI companion called **Sakhi**. The user can ask health questions through text or voice, in the language she is comfortable with. Sakhi first understands the user's concern and emotional context, then provides relevant guidance using a controlled knowledge base built from trusted medical resources such as WHO and NHM materials. It does not freely browse random internet content to generate health answers.

The product has two modes: **Girl Mode** and **Woman Mode**. The selected mode determines which health information and tools are shown on the dashboard. Both modes contain the Sakhi chatbot and Smart Referral Engine.

If Sakhi identifies a potentially high-risk situation, it should not attempt to diagnose the user. Instead, it calmly recommends medical attention and directs the user to the Smart Referral Engine. The referral flow uses risk, location, an appropriate healthcare facility, reliable availability/status information where available, and navigation.

> The product is intentionally focused. It is not designed to become a large hospital-management system, social network, doctor marketplace, or general-purpose AI assistant. The goal is to make a small number of important features feel safe, simple, trustworthy, and genuinely useful.

2. Product Vision

Sakhi aims to become a trusted first place for a girl or woman to talk about a health concern without embarrassment, fear, or judgment.

Vision statement: Give every girl and woman a private, comfortable space where she can talk about her health in her own language, receive trustworthy guidance, understand when a situation may need professional attention, and find an appropriate healthcare facility when necessary.

Core promise: Sakhi listens first. It does not overwhelm the user with medical information. It uses trusted medical knowledge when that knowledge is actually needed and prioritizes safety when risk is detected.

3. Problem Statement

Girls and women may hesitate to discuss menstrual, pregnancy-related, mental wellness, or other health concerns because of embarrassment, fear of judgment, lack of accessible information, language barriers, or uncertainty about whether a symptom is serious.

- Health information found online can be overwhelming, contradictory, or difficult to understand.
- A user may not know whether a symptom is routine, concerning, or urgent.
- Users may prefer speaking naturally rather than filling out long medical forms.
- Many users are more comfortable communicating in their local or preferred language.
- A user who is anxious needs calm communication before detailed information.
- When a situation may be urgent, the user needs a clear next step rather than a long explanation.
- Health information is sensitive, so privacy must be treated as a core product requirement.

4. Product Goals

- Create a simple and friendly women's health experience.
- Make Sakhi conversational, empathetic, non-judgmental, and age-appropriate.

- Support text and voice conversations in multiple languages.
- Provide health guidance grounded in approved, trusted resources rather than random internet knowledge.
- Personalize the dashboard and conversation using the user's selected mode and relevant health context.
- Provide a clear safety path for potentially high-risk situations.
- Protect sensitive health information through privacy-conscious product and technical design.
- Keep the product focused and easy to use rather than adding unnecessary features.

5. Product Scope

In scope: authentication, Girl Mode, Woman Mode, health tracking relevant to each mode, health reports, Sakhi chatbot, multilingual text and voice interaction, controlled medical RAG, risk/safety assessment, Smart Referral Engine, and privacy/security.

Out of scope for the current product: ASHA/doctor/admin dashboards, community analytics, outbreak detection, doctor marketplace, social networking, hospital management, large-scale telemedicine, and unrelated AI features. These are deliberately excluded to keep the product focused.

6. Target Users

User	Description	Primary Needs
Girl	A younger female user using Girl Mode.	Periods, menstrual cycle, weight, mental wellness, reports, private health questions, and referral support when needed.
Woman	An adult female user using Woman Mode.	Periods, pregnancy-related tracking, reports, private health questions, and referral support when needed.

7. User Onboarding & Authentication

The onboarding experience should be intentionally short. The product should not ask for unnecessary information at account creation.

1. User opens Sakhi.
2. User provides her name.
3. User selects Girl or Woman.
4. User creates/provides a password.
5. The account is created and the user is taken to the relevant dashboard.
6. The selected mode controls the dashboard experience.

Requirement: Authentication must use secure password handling. Passwords must never be stored in plain text.

8. Girl Mode Requirements

Girl Mode provides a clean health dashboard focused on the health areas defined for the product.

Module	Purpose	Expected User Action
Menstrual Cycle	Maintain useful cycle information and history.	Add/view relevant cycle information.

Module	Purpose	Expected User Action
Period Tracking	Record and review periods.	Record dates and view history.
Weight	Track weight over time.	Add and view weight records.
Mental Wellness	Give the user a simple place to record wellness/mood information.	Record how she is feeling and review her own history.
Health Reports	Store/view health reports privately.	Upload or view reports.
Sakhi	Ask health questions through natural conversation.	Send text or voice.
Smart Referral	Find appropriate care when needed.	Open referral flow and navigate to a facility.

9. Woman Mode Requirements

Woman Mode provides a similarly simple dashboard, with the health areas specifically defined for women using this mode.

Module	Purpose	Expected User Action
Periods	Record and review period information.	Track relevant dates/history.
Pregnancy	Maintain relevant pregnancy information and history.	Record/view pregnancy-related information.
Health Reports	Store/view health reports privately.	Upload or view reports.
Sakhi	Ask health questions through natural conversation.	Send text or voice.
Smart Referral	Find appropriate care when needed.	Open referral flow and navigate to a facility.

10. Sakhi AI Companion

Sakhi is the heart of the product. The intended experience is not a search engine and not a generic question-answer bot. Sakhi should feel like a friendly, understanding girl/woman of an appropriate age who listens carefully, understands the user's concern and emotions, and then provides appropriate guidance.

10.1 Personality Principles

- **Warm:** Use friendly, natural language.
- **Empathetic:** Acknowledge fear, stress, embarrassment, or confusion when present.
- **Non-judgmental:** Never shame the user for a question or symptom.
- **Calm:** Reduce unnecessary anxiety instead of escalating it.
- **Clear:** Explain health information in simple language.
- **Age-appropriate:** The conversation should feel suitable to the user's selected mode.
- **Safety-aware:** Become calm and serious when potentially urgent symptoms are detected.
- **Honest:** Never claim to be a human doctor or pretend certainty when the information is uncertain.

10.2 Conversation Philosophy

Sakhi should **listen first, understand second, guide third**. The system should not immediately respond with a long medical explanation just because a health topic was mentioned.

Example — anxious user: *"My period is late and I'm really scared."* Sakhi should first acknowledge the anxiety and ask only the relevant follow-up questions. If reliable medical information is required, it can then retrieve the relevant information from the approved knowledge base.

The RAG system works in the background. The user should experience a natural conversation, not a visible "searching the internet" process.

10.3 Input Modes

- Text input.
- Voice input.
- Multiple supported languages.
- The system should detect/understand the user's language and respond in the same language whenever supported.
- Voice input should be converted into meaningful structured understanding, not treated only as raw transcription.

10.4 Context Awareness

Where relevant and permitted, Sakhi can use the user's existing information from the product, such as period history, pregnancy information, wellness records, or previously discussed health context. This allows the user to speak naturally without repeatedly explaining the same background.

11. Medical Knowledge & RAG Requirements

Sakhi must not use unrestricted internet knowledge as the source of medical guidance. Health-related answers should be grounded in a controlled knowledge base made from trusted and approved resources.

11.1 Knowledge Sources

- World Health Organization (WHO) resources.
- National Health Mission (NHM) resources.
- Relevant Ministry of Health and Family Welfare (MoHFW) resources.
- Relevant ICMR resources.
- Other authoritative women's-health guidelines/resources approved by the product team.

11.2 RAG Behavior

1. Understand the user's question and determine whether medical knowledge is required.
2. Retrieve relevant content from the approved knowledge base.
3. Provide the retrieved context to the language model.
4. Generate an answer grounded in that context.
5. Avoid inventing unsupported medical claims.
6. Where practical, show the source/resource used for the answer.

If the knowledge base does not contain enough information to answer safely, Sakhi should say so rather than filling the gap with random web information or fabricated certainty.

11.3 Important Safety Boundary

Sakhi is a health guidance and safety-support system, not a replacement for a qualified healthcare professional. It should not present itself as a doctor, make definitive diagnoses, or prescribe treatment as though it were a clinician.

12. End-to-End Product Workflow

The following workflow describes the intended product behavior from entry to assistance.

Step	User/System Action	Outcome
1. Entry	User opens Sakhi.	Welcoming, simple start.
2. Login	Name + Girl/Woman + password.	Authenticated user enters the correct mode.
3. Dashboard	System shows only the relevant mode's features.	Clean, personalized experience.
4. Concern	User asks Sakhi a question by text or voice.	Sakhi receives natural input.
5. Understanding	System interprets language, intent, emotional context, and relevant user context.	Sakhi understands what the user needs.
6. Guidance	If medical knowledge is required, retrieve approved RAG context.	Grounded guidance in the user's language.
7. Safety	Safety/risk logic checks for potentially urgent signs.	Normal conversation continues or escalation is recommended.
8. Referral	For high-risk cases, user is directed to Smart Referral.	Appropriate facility can be identified.
9. Navigation	User receives facility/location/navigation information based on available data.	User can proceed toward care.

13. Safety & Smart Referral Engine

Smart Referral is available in both Girl Mode and Woman Mode. It is the product's safety path for potentially high-risk situations.

13.1 High-Risk Conversation Behavior

When Sakhi identifies symptoms that may require prompt professional attention, it should not frighten the user or attempt to diagnose her. It should communicate clearly and calmly that the symptoms may be urgent and direct her to Smart Referral.

Example: *"I understand this may feel scary. The symptoms you described may need prompt medical attention. Please open Smart Referral so we can help you find an appropriate healthcare facility near you."*

13.2 Smart Referral Flow

Risk → Location → Nearest appropriate facility → Availability/status → Navigation

- Risk level is passed from the safety layer to the referral flow.
- The system uses the user's location with appropriate permission.
- The system determines the appropriate facility type for the situation.

- Nearby suitable facilities are identified.
- Availability/status is shown only when reliable information is available; the system must not invent availability.
- The user can open navigation to the selected facility.

13.3 Safety Boundary

The risk engine is intended to support triage and escalation, not diagnosis. The UI should avoid statements such as “You have disease X.” It should instead communicate potential urgency and recommend appropriate professional care.

14. Privacy & Security Requirements

Health information is highly sensitive. Privacy is therefore a core product requirement, not a final-stage enhancement.

- Use secure authentication and secure password hashing.
- Do not store passwords in plain text.
- Enforce authorization so a user can access only her own health information.
- Protect health reports and other sensitive files from unauthorized access.
- Use HTTPS/TLS in deployed environments.
- Use secure API authentication and session/token handling.
- Collect only information necessary for the product.
- Do not expose health information through logs, URLs, error messages, or client-side code unnecessarily.
- Do not share health data with third parties without a valid product/legal basis and appropriate user consent.
- Design data deletion and retention behavior intentionally.
- Maintain clear privacy messaging so users understand how their information is handled.

Privacy principle: The safest design is to avoid collecting or exposing information that the product does not actually need.

15. UI/UX Requirements

The product should feel calm, approachable, modern, and uncluttered. The user may open the app while worried or embarrassed, so the interface should reduce cognitive load.

- Simple navigation.
- Clear labels and icons.
- Minimal unnecessary forms.
- Readable typography and strong visual hierarchy.
- Responsive design for mobile-first use.
- Friendly but mature visual language.
- Important actions such as Sakhi and Smart Referral should be easy to find.
- Error messages should be understandable and non-technical.
- Loading states should reassure the user rather than showing confusing technical details.

- Critical/referral states should be visually clear without using frightening language.

16. Key Screens

Screen	Purpose	Core Content
Welcome	Introduce Sakhi and create a calm first impression.	Sakhi identity, short message, start/login action.
Login/Onboarding	Create/access account.	Name, Girl/Woman selection, password.
Girl Dashboard	Central health home for Girl Mode.	Cycle, periods, weight, mental wellness, reports, Sakhi, referral.
Woman Dashboard	Central health home for Woman Mode.	Periods, pregnancy, reports, Sakhi, referral.
Sakhi Chat	Natural text/voice health conversation.	Messages, voice controls, language-aware responses, source references when applicable.
Health Tracking	Record and review health information.	Relevant tracking based on mode.
Reports	Private report access.	Upload/view reports securely.
Smart Referral	Find appropriate care.	Risk context, location, facility, availability/status, navigation.

17. High-Level Technology Direction

The implementation should use technologies that are realistic for the development team and support the product requirements. The team already has working familiarity with HTML, CSS, JavaScript, React, Node.js, Supabase, MongoDB, and GitHub, while Django is known but needs refreshing. The final stack should therefore prioritize maintainability and existing skills rather than technology for its own sake.

Area	Preferred Direction	Reason
Frontend	React + modern CSS/Tailwind if useful	Strong fit for responsive dashboards and reusable UI components.
Backend	Node.js/Express or Django REST after final evaluation	Both are viable; choose one primary backend to avoid unnecessary complexity.
Database/Auth	Supabase/PostgreSQL is a strong candidate	Can simplify authentication, database, storage, and deployment for a focused product.
AI Layer	Python-based AI service/module	Useful ecosystem for RAG, language processing, voice, and safety logic.
RAG	Vector search using a PostgreSQL vector capability or suitable vector store	Supports controlled retrieval from approved medical documents.
Voice	Speech-to-text + text-to-speech services/models	Required for the voice-first conversational experience.
Maps/Referral	Maps/Places provider with reliable location/facility data	Supports facility discovery and navigation.
Version Control	Git + GitHub	Collaboration, versioning, issue tracking, and portfolio visibility.

Technology selection note: The table is a high-level direction, not a command to use every technology listed. The detailed technical design should finalize one coherent stack before implementation.

18. Functional Requirements

ID	Requirement	Priority
FR-01	User can create/login to a private account using the defined onboarding information.	Must
FR-02	User can select Girl Mode or Woman Mode during onboarding.	Must
FR-03	System displays the appropriate dashboard based on selected mode.	Must
FR-04	Girl Mode supports menstrual cycle, period, weight, mental wellness, reports, Sakhi, and Smart Referral.	Must
FR-05	Woman Mode supports periods, pregnancy, reports, Sakhi, and Smart Referral.	Must
FR-06	User can send text messages to Sakhi.	Must
FR-07	User can send voice input to Sakhi.	Must
FR-08	Sakhi responds in the user's language when supported.	Must
FR-09	Health guidance is grounded in the approved medical knowledge base.	Must
FR-10	Sakhi should understand emotional context and adapt tone appropriately.	Must
FR-11	System performs a safety/risk check for potentially urgent symptoms.	Must
FR-12	High-risk cases can be routed to Smart Referral.	Must
FR-13	Smart Referral supports location, appropriate facility selection, reliable availability/status where available, and navigation.	Must
FR-14	User health information is protected through authentication and authorization.	Must

19. Non-Functional Requirements

Area	Requirement
Usability	A first-time user should understand the main actions without training.
Performance	Normal dashboard interactions should feel responsive; AI/voice operations should provide clear loading feedback.
Security	Authentication, authorization, secure storage/access, HTTPS, and secure file handling are required.
Privacy	Only necessary health information should be collected and exposed.
Reliability	Failures in AI, voice, RAG, maps, or facility data should fail safely and clearly.
Accessibility	Readable text, understandable controls, adequate contrast, and voice-friendly interaction should be considered.
Maintainability	Frontend, backend, AI/RAG, and referral logic should have clear boundaries.

20. Important Edge Cases

- User sends an unclear or incomplete health question → Sakhi asks a small number of relevant clarifying questions.
- User switches languages → Sakhi follows the new language when supported.

- Voice transcription is unclear → Sakhi should confirm what it understood rather than confidently inventing details.
- RAG has no relevant source → Sakhi should be transparent and avoid unsupported medical claims.
- Facility availability data is unavailable → do not display a fabricated availability status.
- Location permission is denied → explain that location is needed for nearby referral and offer the appropriate alternative available in the product.
- AI service is temporarily unavailable → show a clear, calm error and avoid pretending that an answer was generated.
- User reports potentially urgent symptoms → prioritize safety and referral over lengthy conversation.
- User uploads an unsupported/corrupted report → show a clear error without exposing sensitive technical information.
- User attempts to access another user's data → deny access and log the security event appropriately.

21. Acceptance Criteria

The initial product can be considered successful when a new user can complete the core journey without external help.

1. Create an account and select Girl or Woman mode.
2. Reach the correct dashboard.
3. Use the relevant health tracking features.
4. Open Sakhi and send a text question.
5. Send a voice question and receive a response when supported.
6. Have a conversation in a supported non-English language and receive the response in the same language.
7. Ask a health question and receive a response grounded in the approved knowledge base.
8. Experience an empathetic response when the message indicates stress or worry.
9. Trigger a high-risk test scenario and receive a clear referral recommendation.
10. Open Smart Referral and proceed through risk → location → appropriate facility → available status information → navigation.
11. Verify that one user cannot access another user's private health information.

22. Testing Strategy

- **UI testing:** Verify dashboard navigation, forms, tracking, reports, chat, and referral screens.
- **API testing:** Verify authentication, authorization, health-data operations, chat endpoints, and referral endpoints.
- **AI testing:** Test common questions, ambiguous questions, emotional messages, unsupported questions, and unsafe/urgent scenarios.
- **RAG evaluation:** Verify that responses are supported by retrieved approved sources and that irrelevant documents are not used.
- **Voice testing:** Test supported languages, accents, noisy environments, unclear speech, and code-switching.

- **Security testing:** Test unauthorized access, insecure file access, token/session handling, and sensitive data exposure.
- **Safety testing:** Create predefined high-risk scenarios and confirm consistent escalation behavior.

23. Product Success Measures

- Users can complete the main flow without assistance.
- Users understand how to start a conversation with Sakhi.
- Sakhi's responses are perceived as supportive and non-judgmental in user testing.
- Health answers remain grounded in approved resources.
- The system does not fabricate medical sources or facility availability.
- High-risk test cases consistently lead to the intended referral flow.
- No unauthorized user can access another user's health information.
- The product remains simple despite its AI and health functionality.

24. Development Approach

Development should proceed incrementally. The team should first establish the product foundation and user experience before adding complex AI behavior.

1. **Foundation:** project setup, authentication, database, navigation, and dashboards.
2. **Health modules:** implement the Girl and Woman dashboard features.
3. **Sakhi interface:** build chat UI and conversation flow before connecting advanced AI.
4. **AI intelligence:** add intent/emotion understanding and controlled response generation.
5. **RAG:** prepare approved resources, retrieval, grounding, and source display.
6. **Voice:** add speech-to-text and text-to-speech with language handling.
7. **Safety:** implement and test risk/escalation logic.
8. **Smart Referral:** integrate location, facility discovery, reliable availability/status information, and navigation.
9. **Security and polish:** harden authentication, files, APIs, privacy, error handling, responsiveness, and deployment.

25. Scope Lock — What Sakhi Is and Is Not

Sakhi IS:

- A private women's health companion.
- A Girl Mode and Woman Mode product.
- A multilingual text and voice conversational assistant.
- A controlled, evidence-grounded health information system.
- A personal health tracking and report interface.
- A safety-aware referral assistant.

Sakhi IS NOT in the current scope:

- A replacement for doctors or emergency services.
- A definitive diagnostic system.
- A prescription system.
- An unrestricted web-search chatbot.
- A doctor/ASHA/admin management platform.
- A social network or community platform.
- A hospital management system.
- A feature-heavy platform with unrelated AI modules.

> This scope lock is intentional. The project should become strong through quality, safety, usability, and thoughtful AI behavior rather than through a large number of unrelated features.

26. Final Product Definition

Sakhi is a private, multilingual women’s health companion that listens like a supportive peer, guides like a carefully grounded health-information system, and escalates safely when professional care may be needed.

A successful Sakhi experience should feel simple on the surface while being thoughtfully engineered underneath. The user should not need to understand RAG, AI models, risk engines, databases, or APIs. She should only need to know that she can open Sakhi, talk naturally, get trustworthy guidance, and find the next appropriate step when something may be serious.

End of Product Requirements Document